1. Scope

This report presents the Day 2 baseline Chidamber & Kemerer (CK) metrics for the eight classes owned by the Desktop GUI and Client Shell work package. These classes collectively implement the non-VR user interface of iDaVIE: file/mask loading, rendering parameter panels, statistics display, source management, paint mode, histogram controls, video recording UI, and tab navigation.

Classes Analysed

Class	File	Role
CanvassDesktop	`Assets/Scripts/UI/CanvassDesktop.cs`	Orchestrator (O)
DesktopPaintController	`Assets/Scripts/UI/DesktopPaintController.cs`	Adapter (A)
PaintMenuController	`Assets/Scripts/Menu/PaintMenuController.cs`	Orchestrator (O)
VideoUiManager	`Assets/Scripts/VideoMaker/VideoUiManager.cs`	Adapter (A)
HistogramMenuController	`Assets/Scripts/Menu/HistogramMenuController.cs`	Adapter (A)
HistogramHelper	`Assets/Scripts/Menu/HistogramHelper.cs`	Domain helper (D)
SourceRow	`Assets/Scripts/Menu/SourceRow.cs`	Domain helper (D)
TabsManager	`Assets/Scripts/Menu/TabsManager.cs`	Domain helper (D)

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2. CK Metrics Summary

Thresholds from the assignment specification (Section 7.1):

- **WMC** ≤ 20 (domain) / ≤ 40 (orchestrator/adapter)
- **DIT** ≤ 4
- **NOC** ≤ 5
- **CBO** ≤ 14 (domain) / ≤ 25 (orchestrator)
- **RFC** ≤ 50
- **LCOM (Henderson-Sellers)** ≤ 0.50

Class	Role	WMC	DIT	NOC	CBO	RFC	LCOM_HS
CanvassDesktop	O	**63** 🔴	1 ✅	0 ✅	**47** 🔴	**118** 🔴	**0.955** 🔴
DesktopPaintController	A	**57** 🔴	1 ✅	0 ✅	21 ✅	**99** 🔴	**0.940** 🔴
PaintMenuController	O	24 ✅	1 ✅	0 ✅	9 ✅	**56** 🔴	**0.919** 🔴
VideoUiManager	A	17 ✅	1 ✅	0 ✅	17 ✅	**64** 🔴	**0.863** 🔴
HistogramMenuController	A	13 ✅	1 ✅	0 ✅	12 ✅	36 ✅	**0.812** 🔴
HistogramHelper	D	3 ✅	1 ✅	0 ✅	13 ⚠	23 ✅	**0.667** 🔴
SourceRow	D	3 ✅	1 ✅	0 ✅	3 ✅	11 ✅	**0.667** 🔴
TabsManager	D	3 ✅	1 ✅	0 ✅	4 ✅	7 ✅	0.467 ✅

🔴 = violation · ⚠ = at/near threshold · ✅ = within threshold

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3. Violation Analysis

3.1 DIT / NOC — No violations

All eight classes sit one level above `MonoBehaviour` (DIT = 1). None have subclasses (NOC = 0). Inheritance is not a concern in this subsystem.

3.2 WMC — 2 violations

CanvassDesktop (63) and **DesktopPaintController (57)** both exceed the orchestrator/adapter ceiling of 40. These are god classes that absorb file-loading, rendering control, statistics display, paint mode, source management, and VR/desktop bridging into a single `MonoBehaviour`.

3.3 CBO — 1 critical violation

CanvassDesktop (47) nearly doubles the orchestrator threshold of 25. It directly couples to:

- 23 project classes (e.g. `VolumeDataSetRenderer`, `FitsReader`, `DataAnalysis`, `FeatureMapping`)
- 13 Unity/TMPro UI types
- 7 System library types (`IntPtr`, `StringBuilder`, `Marshal`, etc.)
- 4 Valve.VR types (`SteamVR`, `OpenVR`)

Any change to `FitsReader`, `VolumeCommandController`, `DataAnalysis`, `FeatureMapping`, or `MenuBarBehaviour` ripples directly into this class.

HistogramHelper (13) is borderline against the domain threshold of 14, driven by its OxyPlot charting dependency.

3.4 RFC — 5 violations

Class	RFC	Threshold	Over by
CanvassDesktop	~118	50	+68
DesktopPaintController	~99	50	+49
VideoUiManager	64	50	+14
PaintMenuController	56	50	+6

High RFC correlates directly with high CBO and WMC — these classes call out to too many external types because they own too many responsibilities.

3.5 LCOM — 7 of 8 classes violate (≤ 0.50)

TabsManager (0.467) is the only clean class.

Critical cases:

- **CanvassDesktop (0.955):** 63 methods, 67 fields, but total field–method access count is only 189. Methods operate on small, largely disjoint subsets of the field set — the textbook signature of a class that should be split. Three fields (`_restFrequency`, `inPaintMode`, `_tabsManager`) are declared but never accessed by any method.
- **DesktopPaintController (0.940):** 57 methods, 66 fields, access count 226. Same disjoint-access pattern. The `axis` field is touched by 20 methods while `firstEnable`, `colormapHeight`, and `minZoom` are dead.
- **PaintMenuController (0.919):** Deceptively bad. `_volumeInputController` is accessed by 16 of 24 methods, but 5 fields (`cropstatus`, `featureStatus`,

oldSaveText, paintMenu, savePopup) are declared and never touched — dead weight inflating LCOM.

- **VideoUiManager (0.863):** _isPaused is declared but never accessed.

Low-priority cases:

- **HistogramHelper (0.667)** and **SourceRow (0.667):** Trivially small classes. Their LCOM violations are artefacts of empty Unity lifecycle stubs (Start/Update with no bodies) and publicly-set data fields that the class never reads internally. These inflate LCOM mathematically but do not indicate a design problem.

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4. Dead Code Inventory

Class	Dead Fields
CanvassDesktop	_restFrequency, inPaintMode, _tabsManager
DesktopPaintController	firstEnable, colormapHeight, minZoom
PaintMenuController	cropstatus, featureStatus, oldSaveText, paintMenu, savePopup
VideoUiManager	_isPaused
HistogramMenuController	editMinScale, editMaxScale

12 dead fields across 5 classes. These should be removed in the refactoring proposal.

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5. Key Coupling Dependencies (CanvassDesktop)

CanvassDesktop's 47-type coupling breaks down as:

Category	Count	Examples
Project types	23	VolumeDataSetRenderer, VolumeInputController, FitsReader, DataAnalysis, FeatureSetManager, FeatureMapping, Config, ColorMapUtils
Unity / TMPro	13	TMP_Dropdown, TMP_InputField, Toggle, Slider, Button, Coroutine, PlayerPrefs

System	7	`IntPtr`, `StringBuilder`, `FileInfo`, `Marshal`, `CultureInfo`
Third-party	4	`StandaloneFileBrowser`, `ExtensionFilter`, `SteamVR`, `OpenVR`

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6. Implications for Refactoring

The two primary refactoring targets are **CanvassDesktop** and **DesktopPaintController**. Both are god classes with extreme WMC, RFC, and LCOM violations.

The assignment specification (Section 6.6) directs the following splits:

1. **CanvassDesktop → MVVM decomposition:** View (Unity 6 UI Toolkit), ViewModel (pure C#), Service Gateway (server communication). Menu structure, panel state, file dialogs, and configuration become separate, composable responsibilities.
2. **File tab:** From direct native-plugin call to ViewModel command via service gateway (Worked Example 1).
3. **Debug tab:** From inline GUI logic to Observer of a structured logging stream (Worked Example 2).

The projected Day 13 metrics for these refactored classes should bring WMC, CBO, RFC, and LCOM within their respective thresholds.

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